

Topic 2: Spatial data handling

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Spatial query languages

Relationships between spatial
objects

Spatial operations

Spatial queries

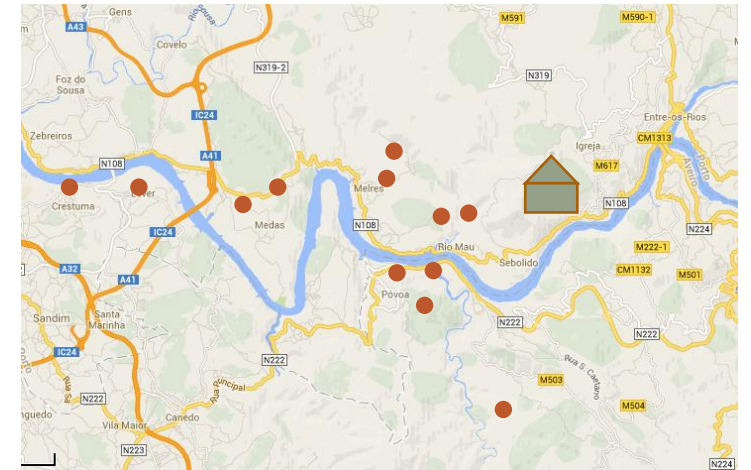
Spatial query languages

spatial operations enable obtaining important information that cannot be unveiled using non-spatial operations

disease spread

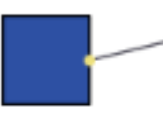


pollution spot



Topological relationships

The point-set approach is used to define topological relationships

	poly-poly	line-line	point-point	poly-line	poly-point	line-point
Disjoint						
Meet						
Overlap						
Contains						

...

Source: http://gitta.info/SpatialQueries/en/html/TopoBasedOps_learningObject1.html

The Point-set approach

It is a formalism to define the relationships between spatial objects

- the 4-intersection model (2x2 matrix: boundary & interior)

the binary topological relationships between two objects, A and B , are defined in terms of the four intersections of A 's boundary and interior with the boundary and interior of B .

$$\begin{bmatrix} A^b \cap B^b & A^b \cap B^i \\ A^i \cap B^b & A^i \cap B^i \end{bmatrix}$$

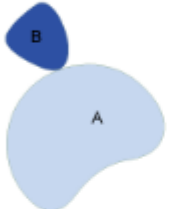
- The 9-intersection model (3x3 matrix : boundary, interior & exterior)

$$\begin{bmatrix} A^b \cap B^b & A^b \cap B^i & A^b \cap B^e \\ A^i \cap B^b & A^i \cap B^i & A^i \cap B^e \\ A^e \cap B^b & A^e \cap B^i & A^e \cap B^e \end{bmatrix}$$

Topological relationships

Source:

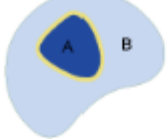
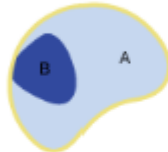
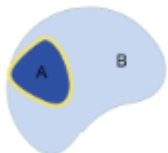
http://gitta.info/SpatialQueries/en/html/TopoBasedOps_1_earningObject1.html

Topological relation	Graphical description	4-intersection-matrix	9-intersection-matrix
Disjoint		$\begin{array}{c} \text{Objekt B} \\ b \quad i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \\ i \end{array}$	$\begin{array}{c} \text{Objekt B} \\ b \quad i \quad e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ e \end{array}$
Meet		$\begin{array}{c} \text{Objekt B} \\ b \quad i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \\ i \end{array}$	$\begin{array}{c} \text{Objekt B} \\ b \quad i \quad e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ e \end{array}$
Overlap		$\begin{array}{c} \text{Objekt B} \\ b \quad i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \\ i \end{array}$	$\begin{array}{c} \text{Objekt B} \\ b \quad i \quad e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ e \end{array}$
Contains		$\begin{array}{c} \text{Objekt B} \\ b \quad i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \\ i \end{array}$	$\begin{array}{c} \text{Objekt B} \\ b \quad i \quad e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \\ e \end{array}$

Topological relationships

Source:

http://gitta.info/SpatialQueries/en/html/TopoBasedOps_1_earningObject1.html

Topological relation	Graphical description	4-intersection-matrix	9-intersection-matrix
Inside		$\begin{array}{c} \text{Objekt B} \\ \begin{array}{cc} b & i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 0 & 1 \\ i & 0 \end{bmatrix} \end{array} \end{array}$	$\begin{array}{c} \text{Objekt B} \\ \begin{array}{ccc} b & i & e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{array} \end{array}$
Covers		$\begin{array}{c} \text{Objekt B} \\ \begin{array}{cc} b & i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \end{array} \end{array}$	$\begin{array}{c} \text{Objekt B} \\ \begin{array}{ccc} b & i & e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \end{array}$
Covered by		$\begin{array}{c} \text{Objekt B} \\ \begin{array}{cc} b & i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \end{array} \end{array}$	$\begin{array}{c} \text{Objekt B} \\ \begin{array}{ccc} b & i & e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \end{array}$
Equal		$\begin{array}{c} \text{Objekt B} \\ \begin{array}{cc} b & i \\ \text{Objekt A} \begin{bmatrix} b & i \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \end{array} \end{array}$	$\begin{array}{c} \text{Objekt B} \\ \begin{array}{ccc} b & i & e \\ \text{Objekt A} \begin{bmatrix} b & i & e \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \end{array}$

Spatial operations

Unary operations

Input is a geometry

- convex
- length, perimeter, area
- rotation, translation, scaling,
simplify (e. g., Douglas-Peucker algorithm)

output is...

→ boolean

→ number

→ geometry

N-ary operations

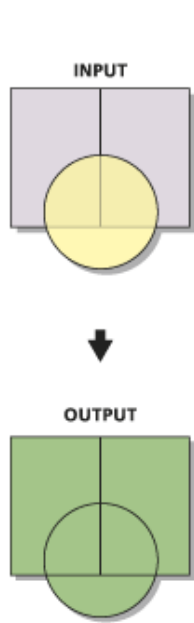
Input is a set (e.g., a set of points) and output is a spatial object (geometry):

- Voronoi Diagram : decomposition of space from a given set of points
- Convex-hull : the convex-hull of a set X of points in the Euclidean space is the smallest convex set that contains X

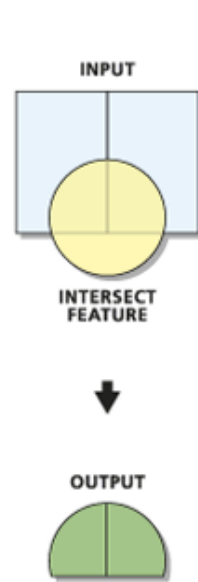
Spatial operations

Operations to compute new spatial objects

- Intersection, Union Difference, Clipping, Dissolve, Merge, Buffer



Union



Intersect



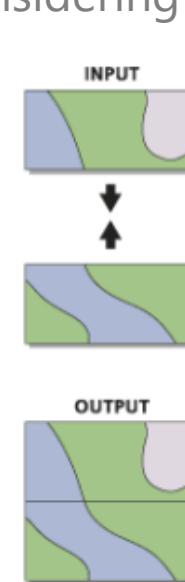
Identity



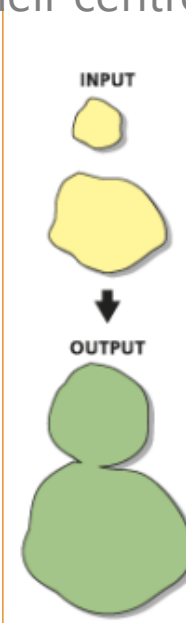
Clip



Dissolve



Merge



Buffer

Operations returning a scalar

- distance, considering the closest points (computational intensive) or considering their centroids

Spatial predicates

Operations that return *true* or *false*

- Topological relationships
- Directional relationships
- Metric relationships

Spatial operations and predicates in PostGIS

- https://postgis.net/docs/using_postgis_query.html
- https://postgis.net/workshops/postgis-intro/spatial_relationships.html

Common spatial queries

Spatial selection (or spatial range query) retrieves the objects satisfying a spatial predicate with a well-defined spatial region or object, e.g. find all objects within a query rectangle

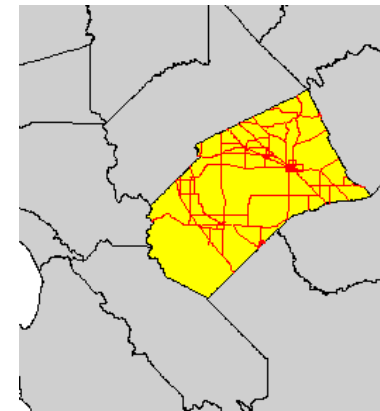
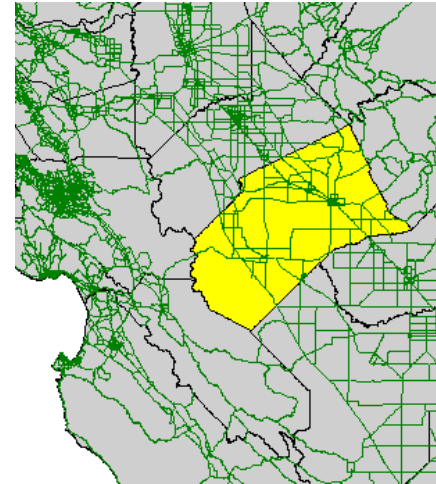
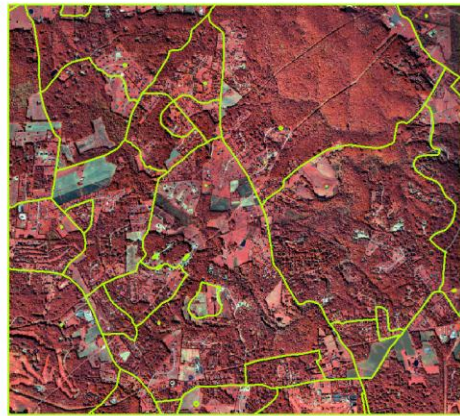
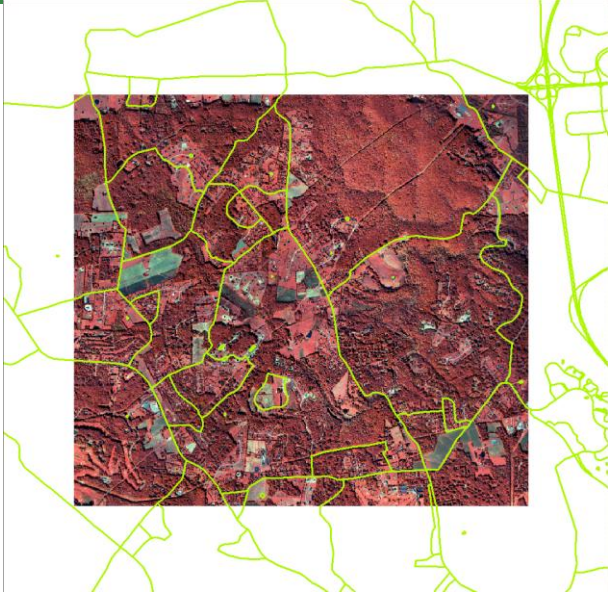
A **point query** is a database operation to retrieve the objects that contain a given point

A **K-nn query** retrieves the top k nearest neighbors to a given object

A **spatial join** combines two relations and retrieves the subset of their Cartesian product that qualifies a spatial predicate (i.e., a spatial relationship)

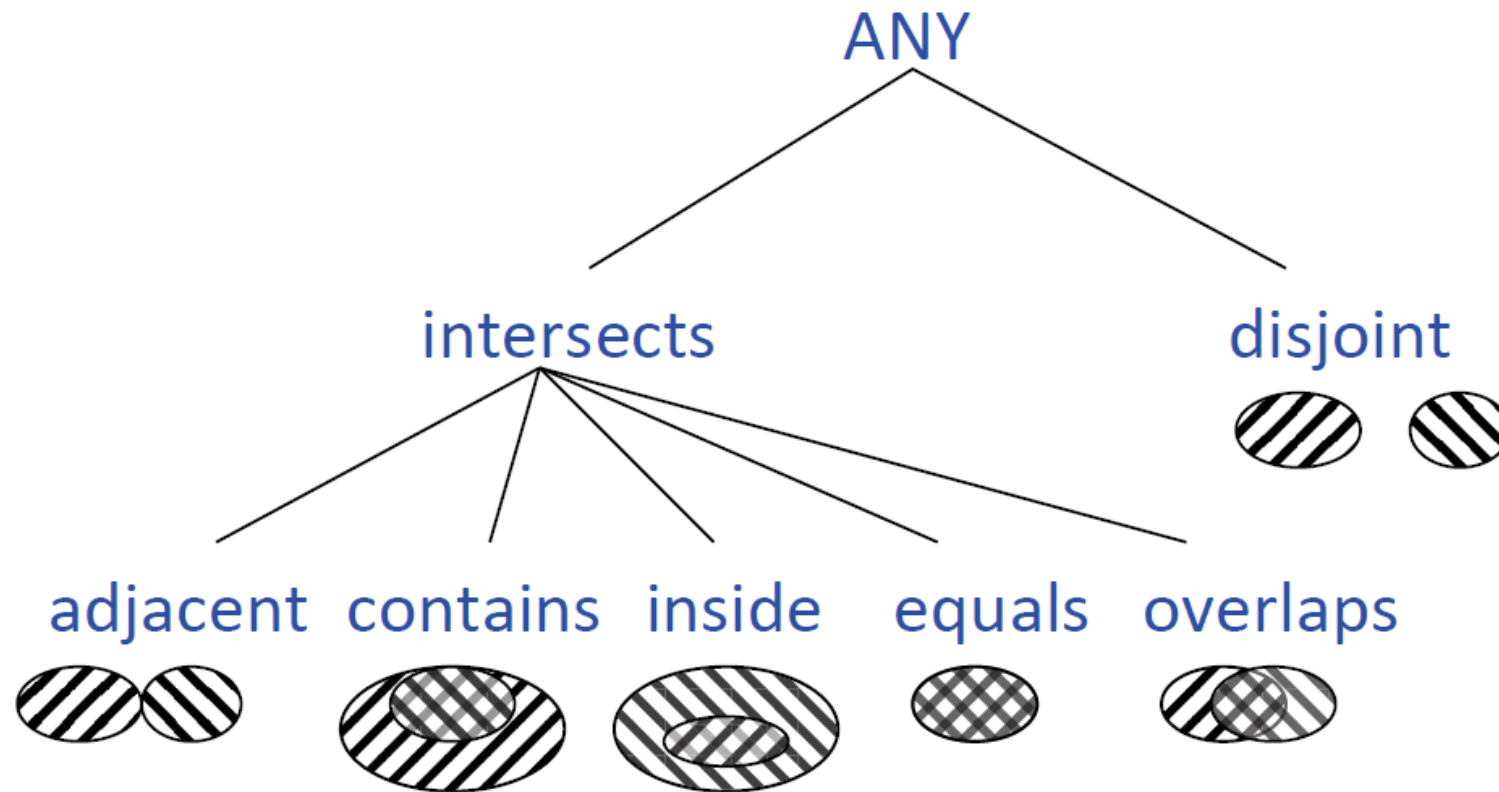
Overlay is a spatial operation that combines different geographic layers to generate new information. Overlay is done using Arithmetic, Boolean, and Relational operators, and is performed in both vector and raster domain

Window and polygon clipping



Topological relationships

The point-set approach is used to define topological relationships



Spatial data formats

- WKT: Well-Known Text (OGC)
- WKB: Well-Known binary (OGC)
- KML: Keyhole Markup Language (KML)
 - XML notation for expressing geographic annotation and visualization within Internet-based, two-dimensional maps and three-dimensional Earth browsers
 - Developed for use with Google Earth
- TIGER: Topologically Integrated Geographic Encoding and Referencing, used by USA Census
- GeoJSON
 - an open format designed for representing simple geographical features, along with their non-spatial attributes, based on JavaScript Object Notation (JSON)
- TopoJSON:
 - an extension of GeoJSON
 - encodes the topology of the objects (topological model) rather than using the spaghetti model

Spatial data formats (shapefile)

Is possibly the most popular geospatial vector data and is developed by ESRI

The format consists of a collection of files with a common filename prefix stored in the same directory

Mandatory files

- .shp: shape format (feature geometry)
- .shx: index of feature geometry (non spatial);
- .dbf: dBASE table that stores the attribute information of features

Other files

- .prj: coordinate system and projection information in WKT
- .sbn and .sbx: a spatial index of the features
- .cpg: code page (only for .dbf) identifying the character encoding to be used
- .qix: an alternative quadtree spatial index used by MapServer and GDAL/OGR software

*ESRI. ESRI Shapefile Technical Description
An ESRI white paper, 1998.*

<https://www.esri.com/library/whitepapers/pdfs/sha>

Spatial query processing

Space filing curves

Data structures and access methods

Algorithmic strategies

What's new ?

Non-spatial (e.g., relational DBMS) databases

- Data
 - Data types are number, string, date or currency
 - one-dimensional (totally ordered)
- Operations:
 - search on one-dimensional totally order data types
 - CRUD: select, insert, update and delete

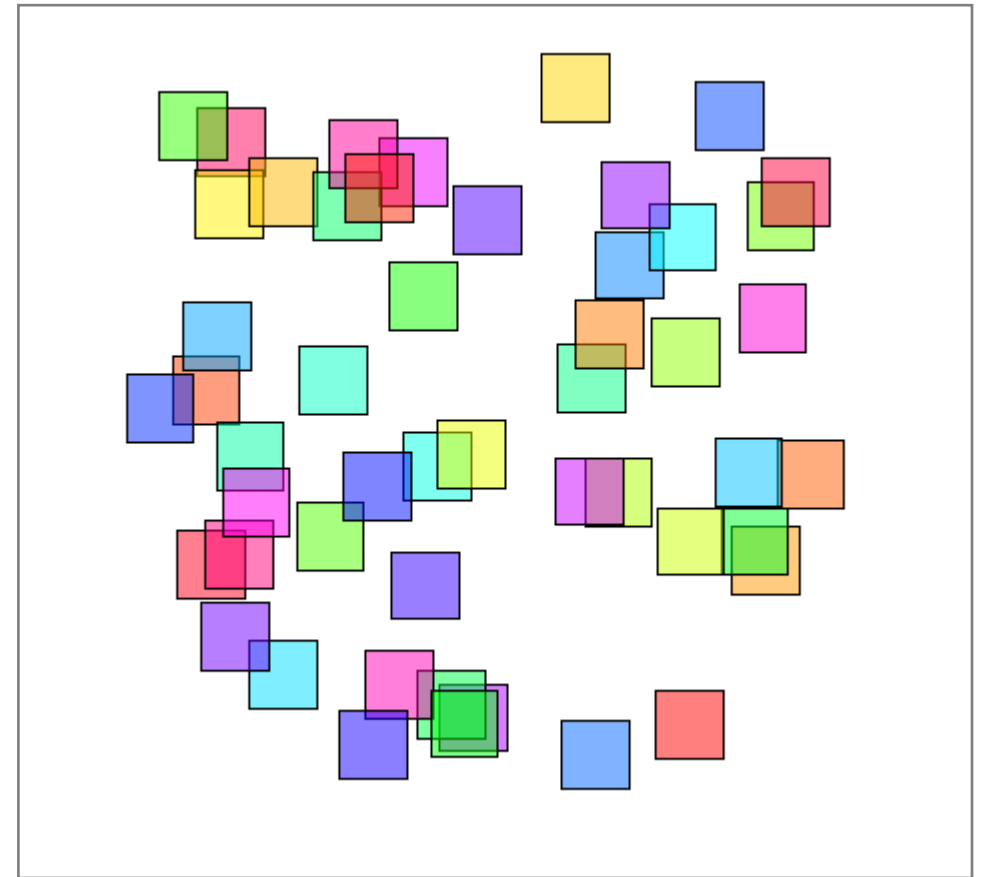
Spatial data is not one-dimensional (it is only partially ordered)

- Space filling curves define a total order for points in 2 or higher dimensional spaces allowing using ordered files
- But we also need new computationally efficient strategies, such as spatial indexes for spatial data analysis

Ordering in multidimensional spaces

How to sort an arbitrary set of 2D points ?

- it does not exist a total/linear order: objects that are close in space may not be close in the total order
- but it is desirable to store objects that are close to each other in space in the same disk page or in the same node in a cluster

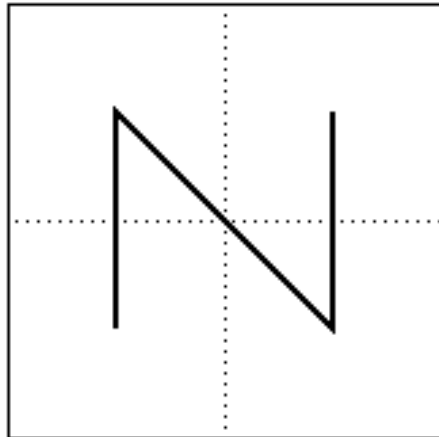


Space filing curve : Z-order curve

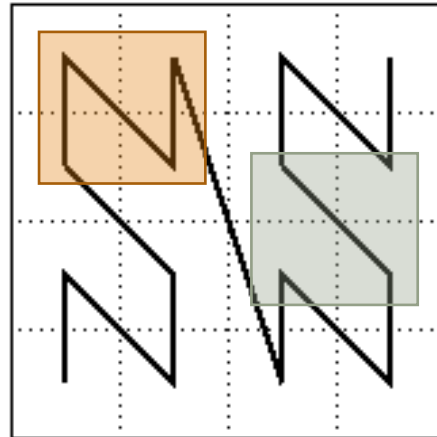
Defines a mapping from a 2D domain to a one-dimensional domain

- A point (cell) in space is mapped into a unique number
- The curve defines a linear ordering of the possible points in the 2D domain
- The linear ordering enables using popular indexes such as B-trees

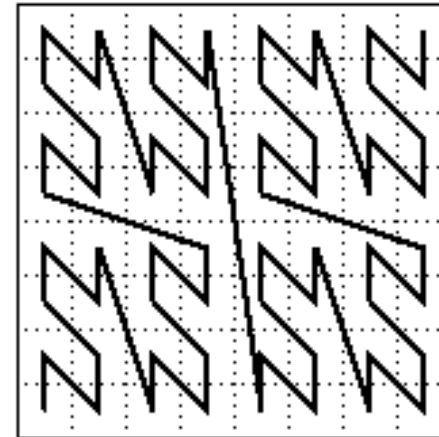
First Order



Second Order



Third Order

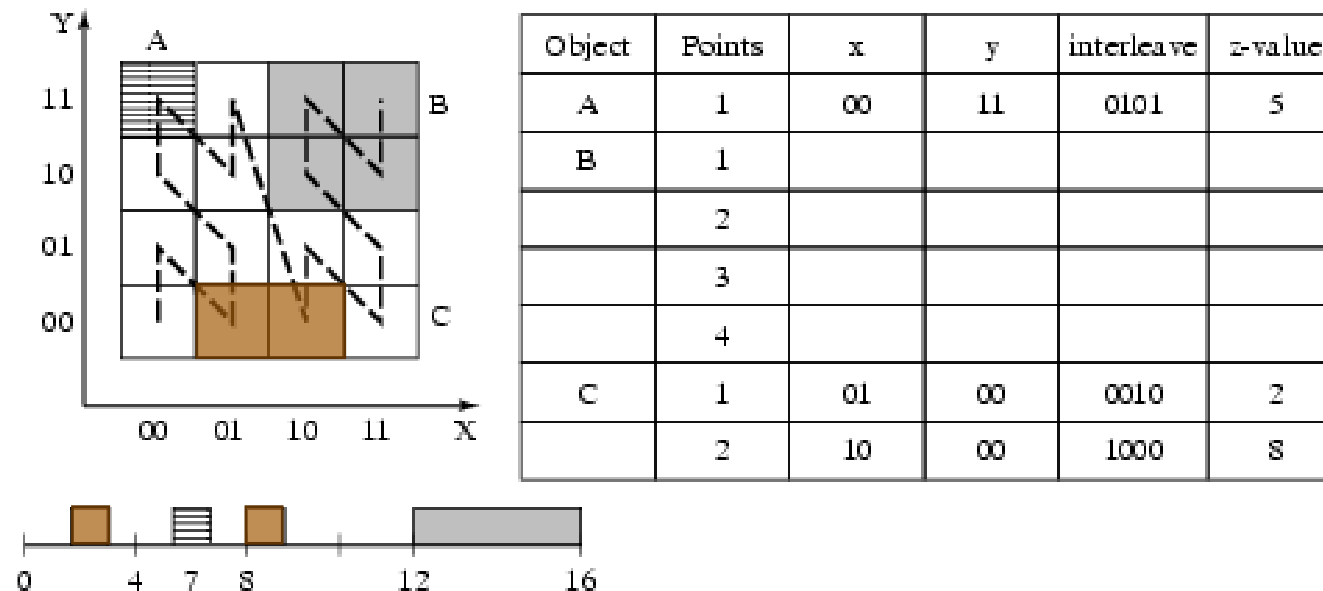


source: <http://www.dcs.bbk.ac.uk/~jkl/BNCOD2000/slides.html>

Example of Z-values

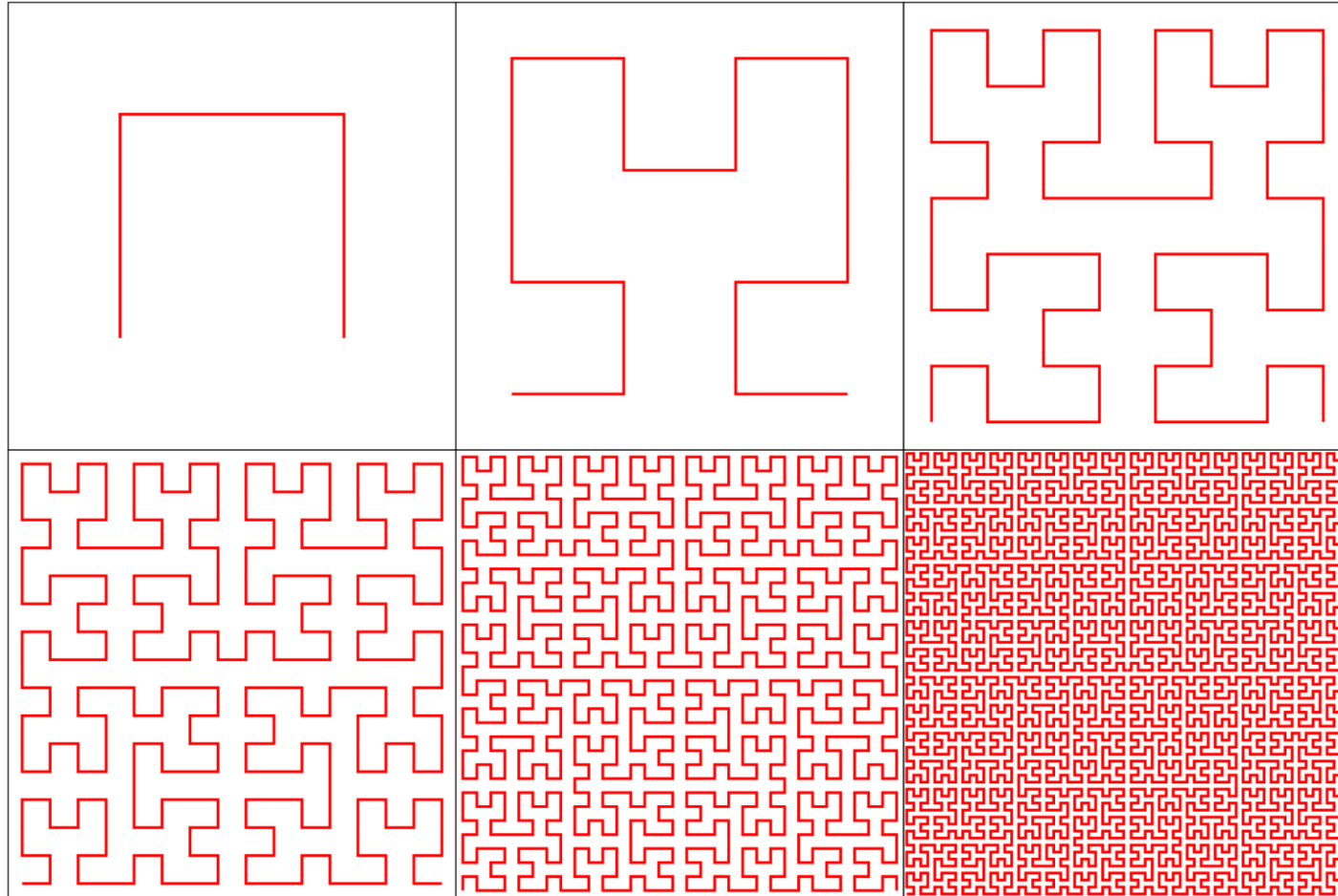
A map with spatial objects A, B, C (left) and the corresponding Z-values of A, B and C (right)

C gets z-values of 2 and 8, which are not close...



Source: Shashi Shekhar and Sanjay Chawla (2003). *Spatial Databases: A Tour*. Pearson.

Space filling curve : Hilbert-curve



source: https://en.wikipedia.org/wiki/Space-filling_curve

Geometry of spatial objects

The manipulation of spatial data is complex due to the complexity of geometries representing the spatial objects

The extent of spatial objects in some applications can be very small compared to the space where they exist and may play a negligible role to the query result

- e.g., in a location-based query 'find the nearest restaurants to...' it is only necessary to know the coordinates of the desired location and the data objects (restaurant) as points

... the extent of the objects can be disregarded in many applications

- Storage, indexing and search can be greatly simplified

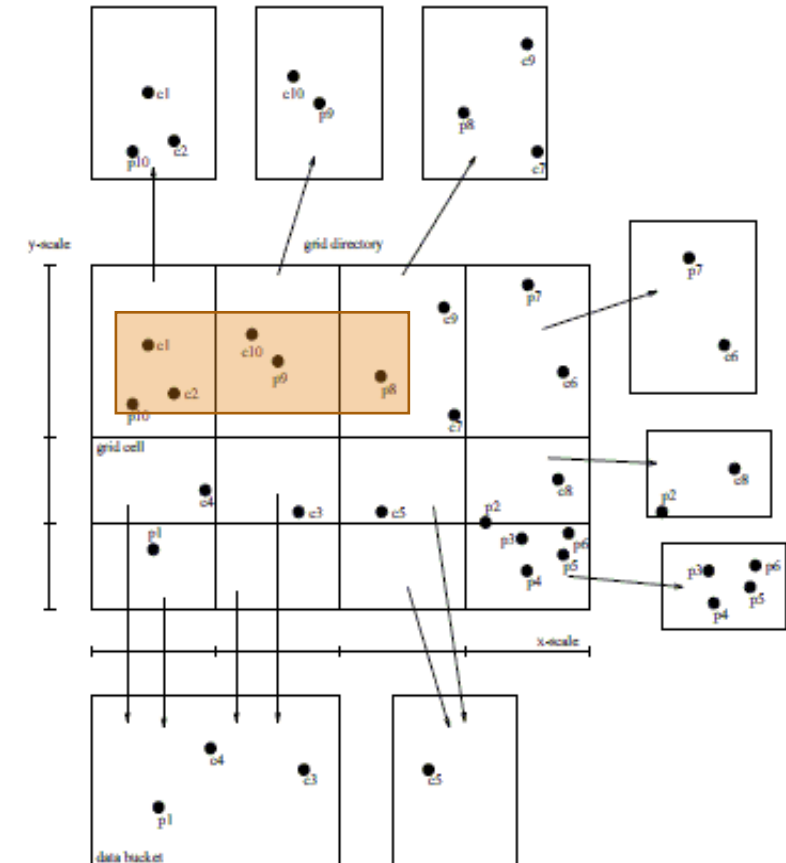
Point access methods : grid file

A grid file divides space into cell using axis-parallel hyper-planes

Consecutive cells are grouped into the same disk page

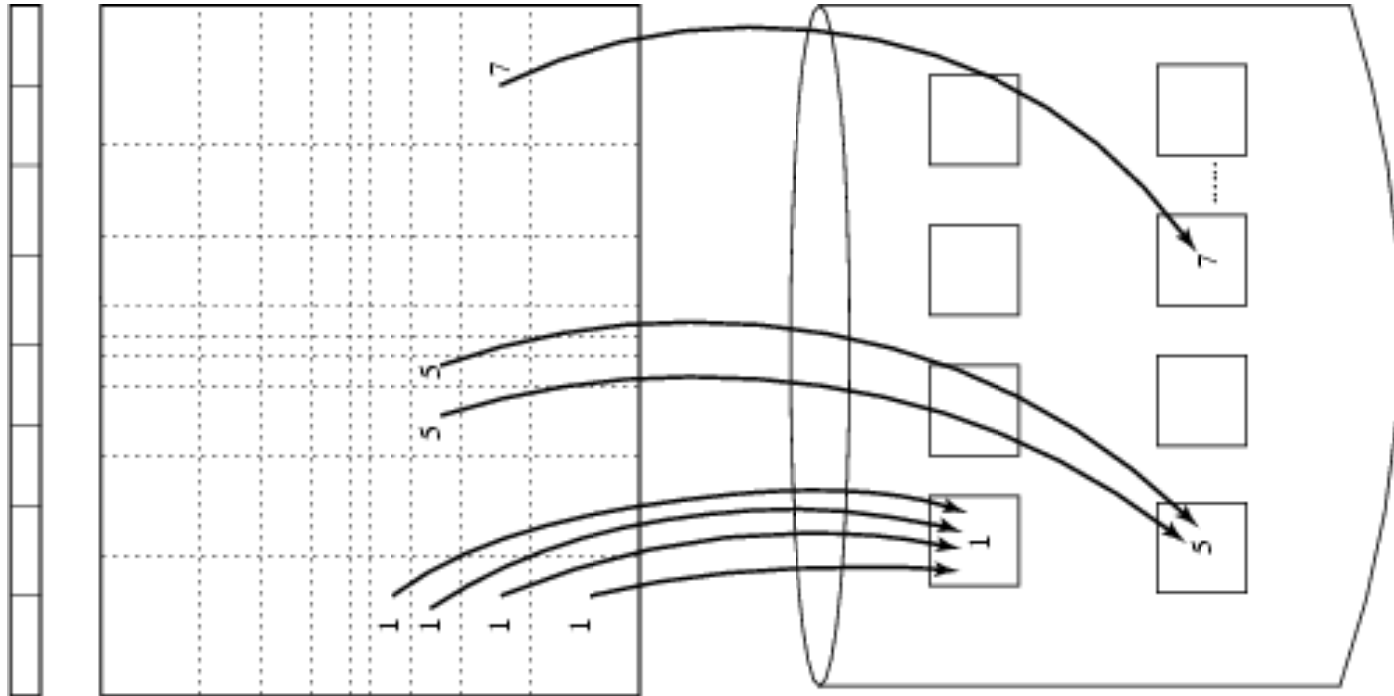
Window queries

- select cells that intersect the query window
- fetch the disk pages that correspond to these cells
- the points in these pages (cell) are compared with the query window
- Good solution when the points are uniformly distributed; for skewed distributions, pages corresponding to sparse areas may be under-utilized



source: Volker Gaede and Oliver Günther. 1998. Multidimensional access methods. ACM Comput. Surv. 30, 2 (June 1998), 170-231.

Point access methods: grid files



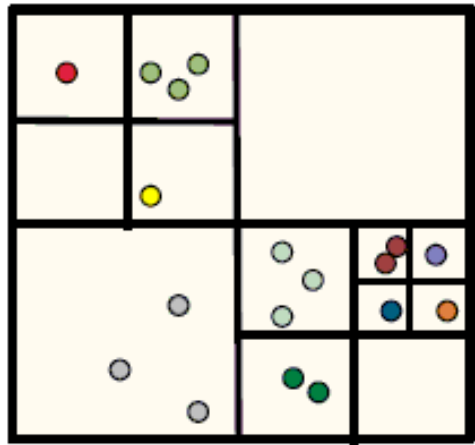
Source: Shashi Shekhar and Sanjay Chawla (2003). *Spatial Databases: A Tour*. Pearson.

Point access methods: quadtree

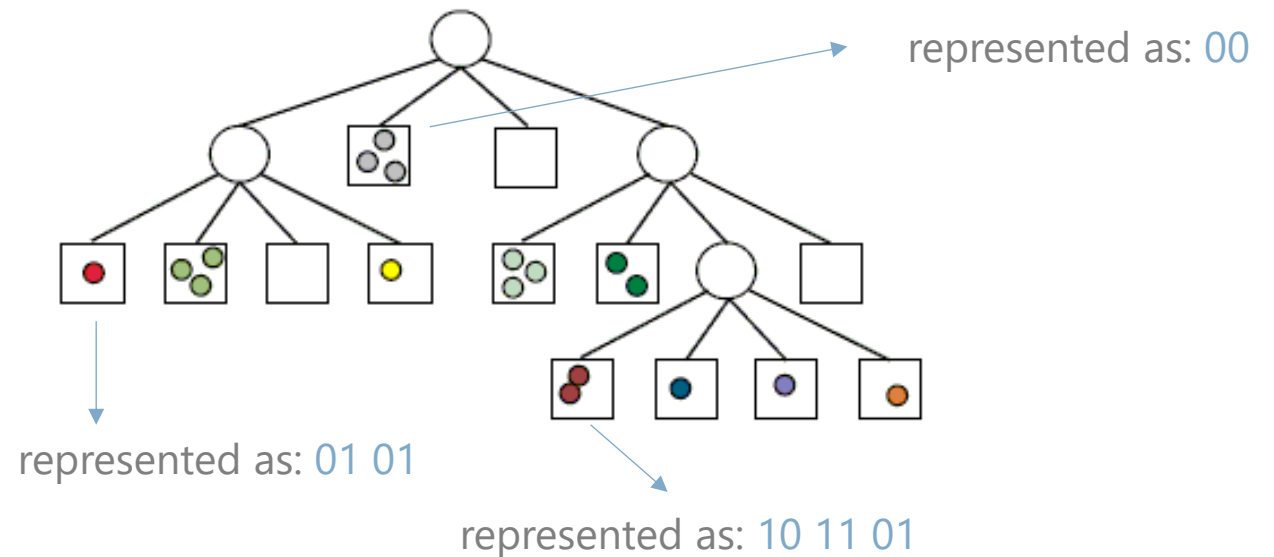
Space is divided recursively into quadrants

Leaf nodes have no more points than the capacity of a disk page

Not necessarily balanced



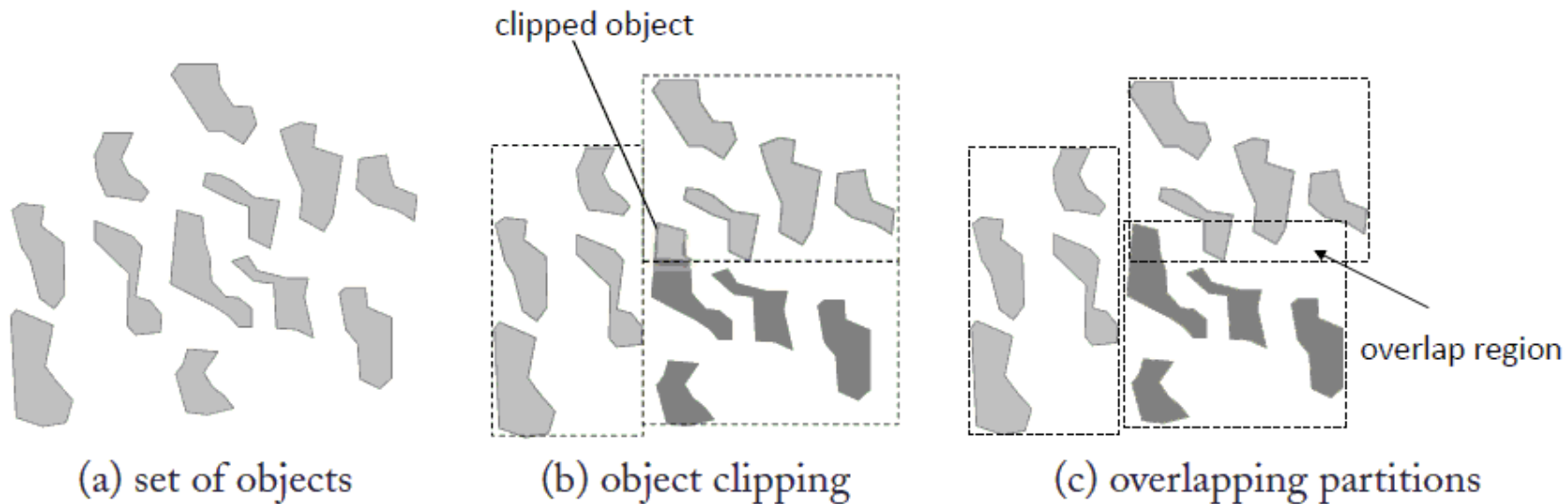
A regular decomposition of space may not adapt well to the distribution of points in space and many leaf nodes may be under-utilized



Indexing objects with extent

Point access methods define space decompositions using disjoint partitions... but a spatial object with extent may be in several cells or quadrants

- Object clipping leads to replication and redundancy...
- Replication can be avoided by decomposing the space into regions that are not necessarily disjoint



Minimum Bounding Rectangle (MBR)

The complex geometry of the objects renders the evaluation of spatial queries expensive

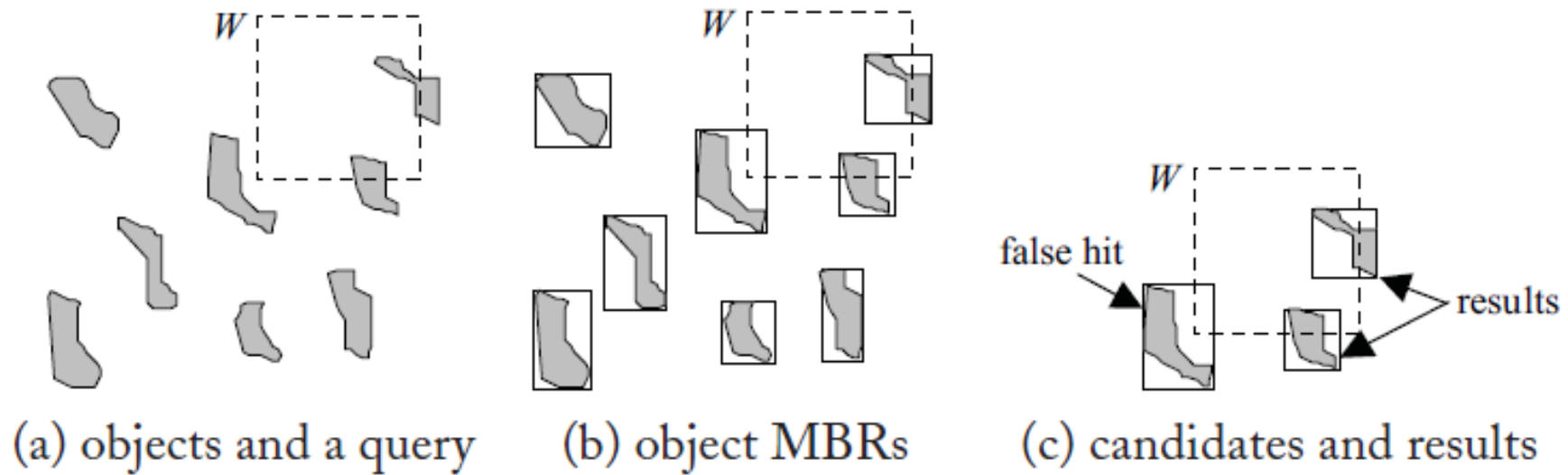
It is possible to use conservative approximations to reduce the computational overhead of spatial queries

- Minimum Bounding Rectangle (MBR) or Minimum Bounding Box (MBB)

Spatial queries are processed in two steps:

- Filter step: the query is applied on MBRs to prune the search space; the result is a set of candidates
- Refinement step: the exact representations of the candidates are tested to retrieve the actual results

Minimum Bounding Rectangle (MBR)



source: Nikos Mamoulis. *Spatial data management. Synthesis Lectures on Data Management*, Morgan&Claypool publishers, 2012.

R-tree

R-Tree is the most popular Spatial Access Method (SAM) and it has been implemented into several Spatial DBMS

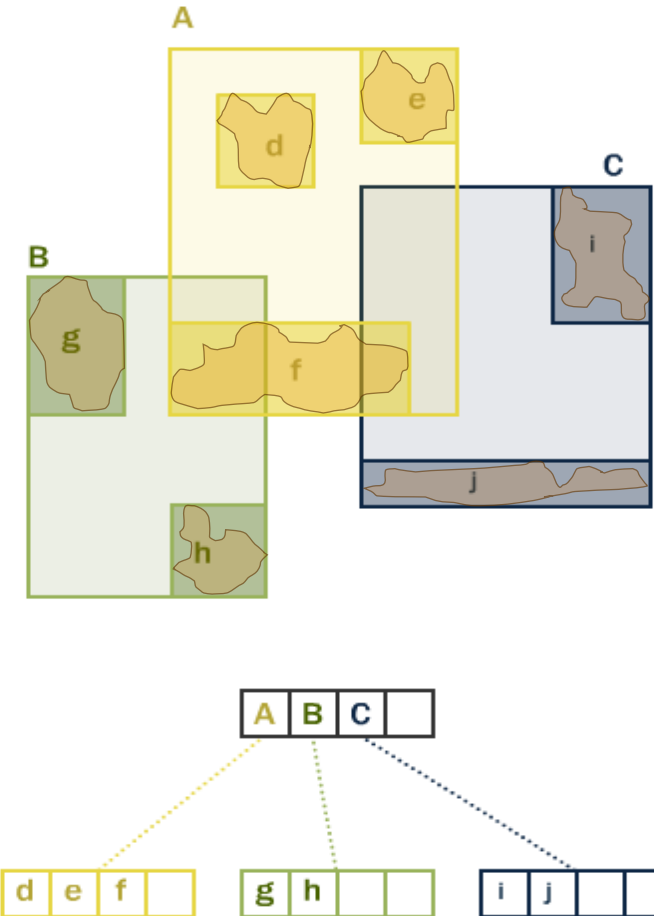
- Can index points, lines and polygons
- Indexes MBRs instead of the exact geometry of spatial objects, like most SAMs
- It is a height-balanced tree similar to a B-tree
- Each R-tree node corresponds to a disk page holding a sequence of entries in the form (MBR, ptr)

R-tree

R-tree node (MBR, ptr)

- Leaf node: MBR is the minimum bounding rectangle of a spatial object and ptr is the id of the object
- Internal node : MBR is the minimum bounding rectangle of all objects pointed by ptr

The number of entries per node depends on the page size

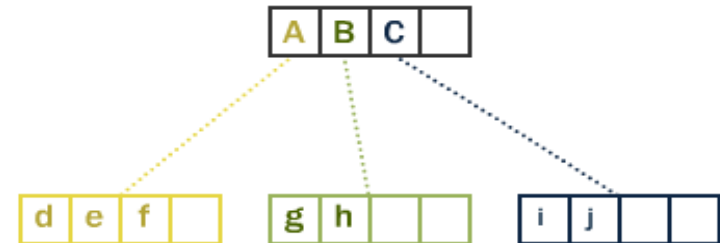
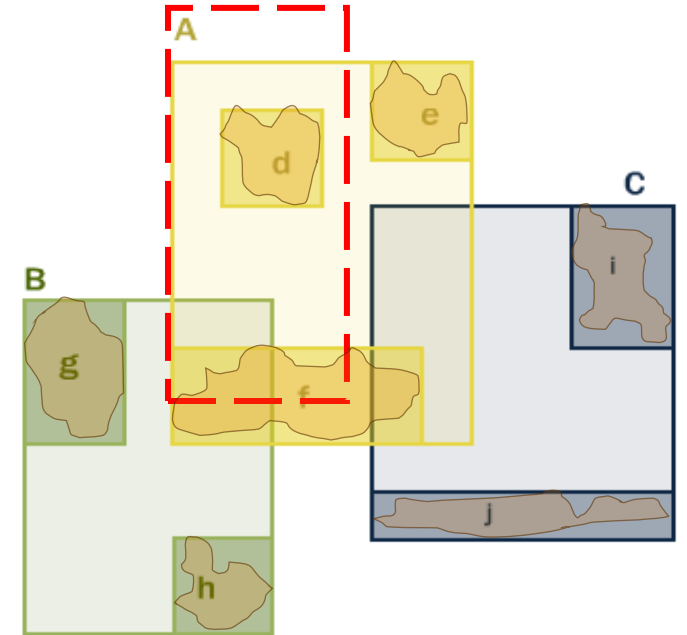


R-Tree : window query

Entry C intersects the query window but none of its objects is in the response set

A good insertion algorithm should minimize dead space and overlapping between nodes

note: dead space is space not occupied by any MBR



Optimization of the R-Tree structure

The area of the MBRs of internal nodes should be minimized

The overlap between MBRs of internal nodes at the same level should be minimized

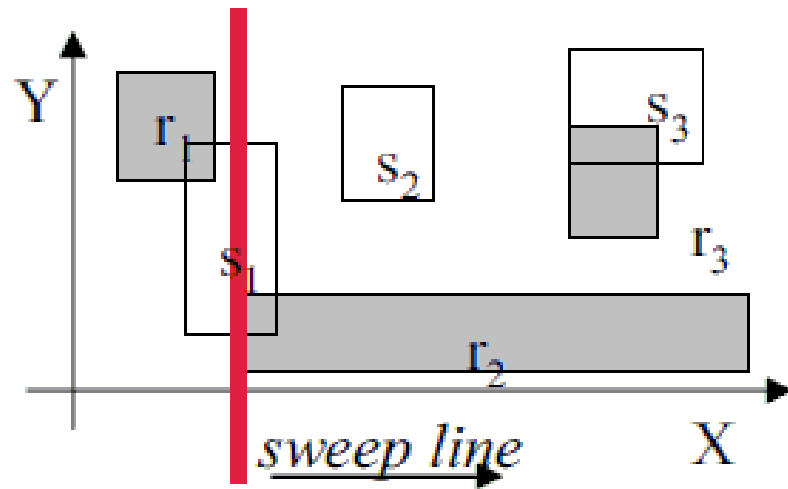
The perimeter of the MBRs of internal nodes should be minimized

The number of entries in the R-tree nodes should be maximized

= = => several versions of R-trees namely, R^+ -tree, R^* -tree

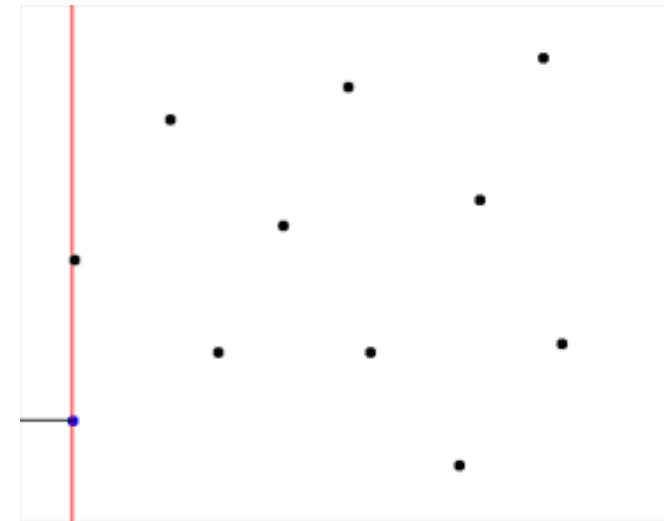
Algorithmic strategies : sweep-line

Decomposition of a geometry into vertical planes and perform the computations in the order that information is encountered



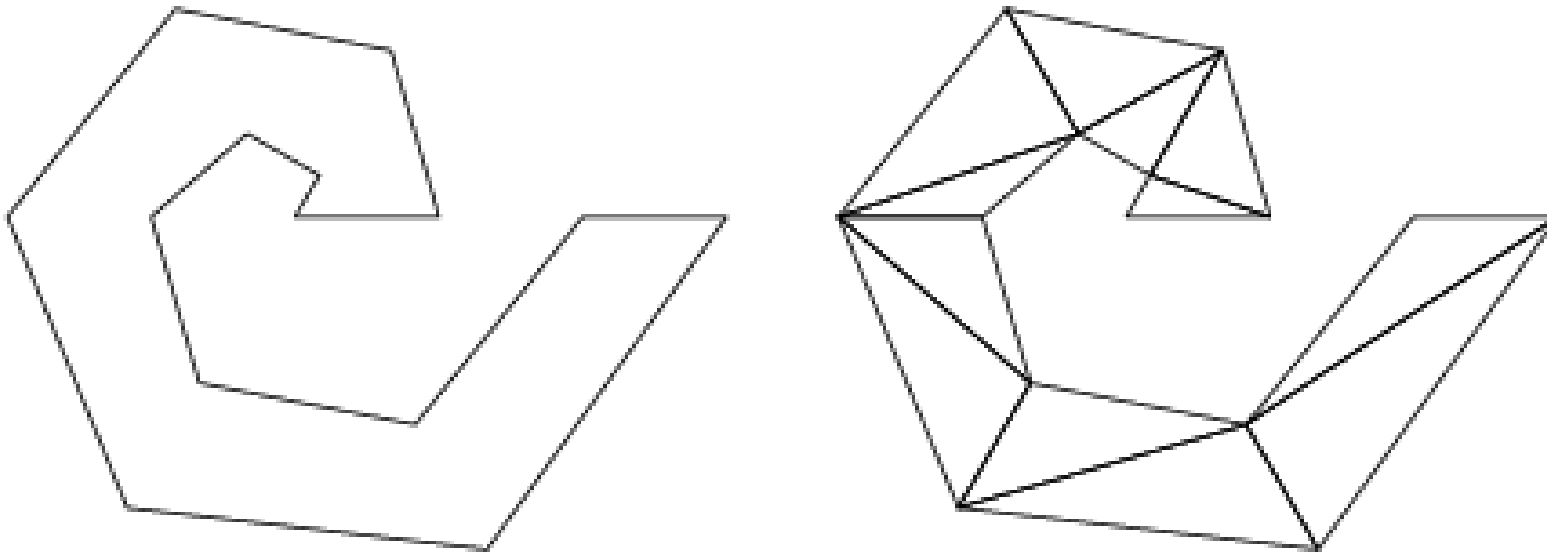
Can be more complex geometries than rectangles

Many applications, e.g., to retrieve the pairwise intersections of a set of rectangles or to compute a Voronoi Diagram



Algorithmic strategies : polygon partitioning

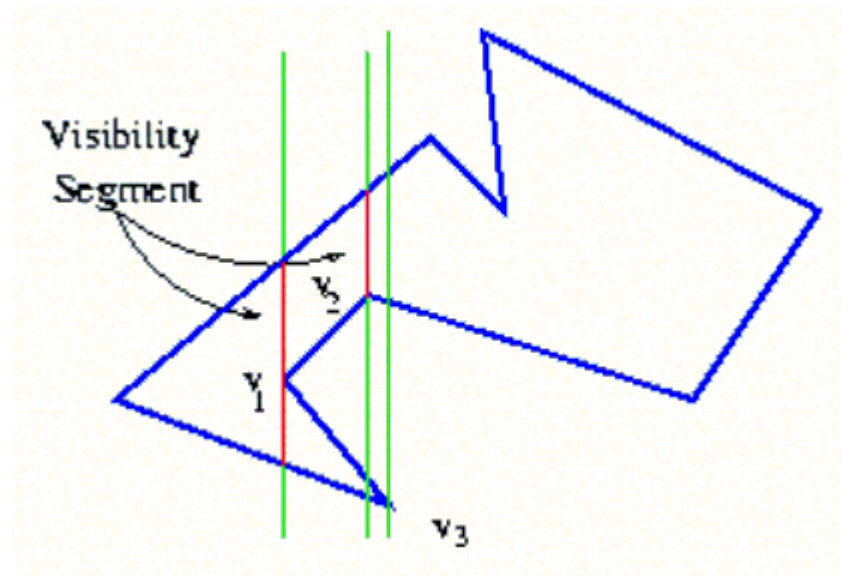
Triangulation of a polygon: these methods are non-deterministic



source: <http://mathworld.wolfram.com/Triangulation.html>

Algorithmic strategies : polygon partitioning

Trapezoidalization of a polygon : relies on sweep-line technique



source: <http://www.personal.kent.edu/~rmuhamma/Compgeometry/MyCG/PolyPart/polyPartition.htm>

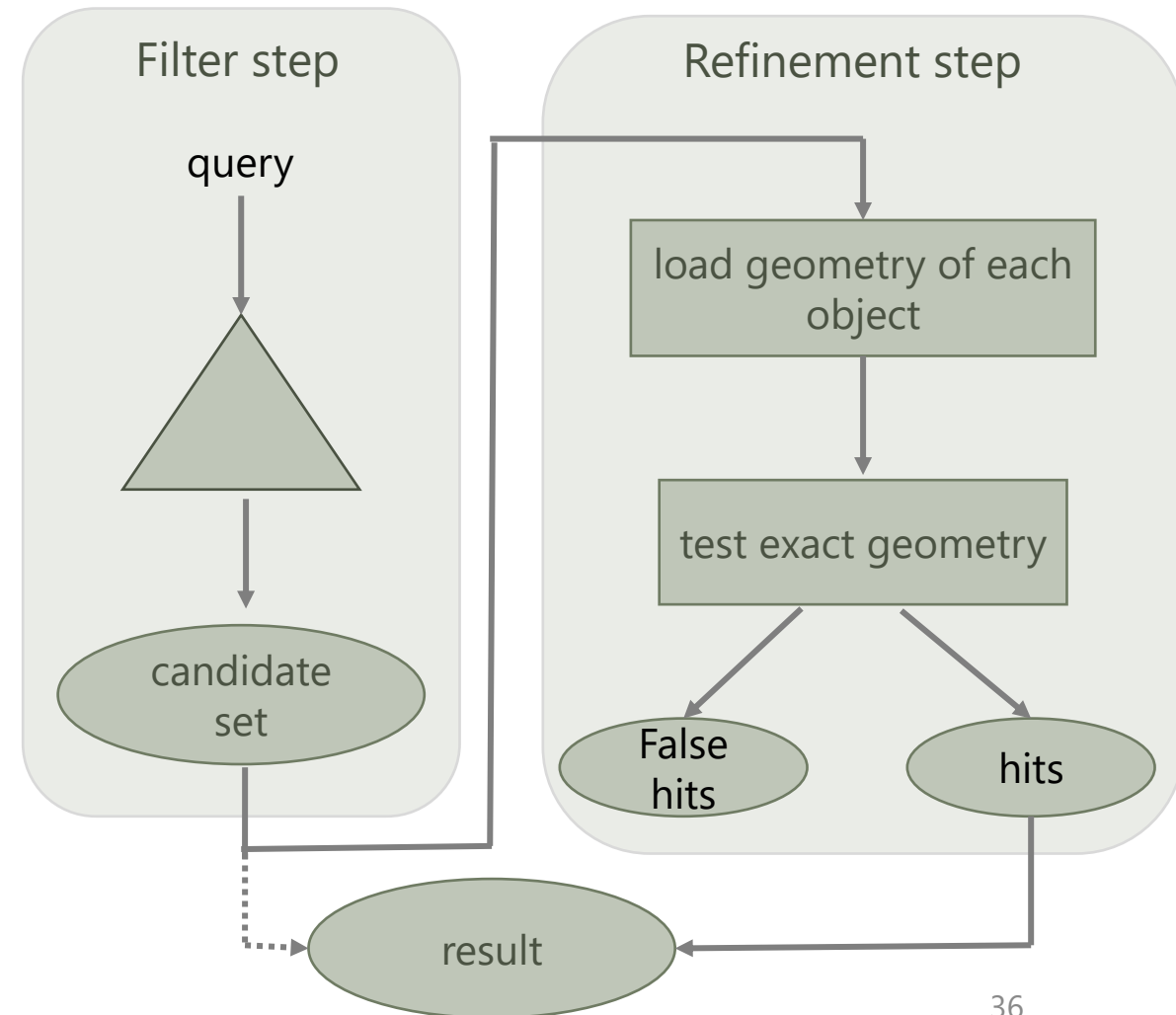
Spatial query processing optimization

Cost models are more complex since

- Spatial queries are both CPU and I/O intensive
- while traditional queries are mainly I/O intensive

Queries evaluated in two steps

- Filter step to find qualifying candidates
- Refinement step to find the exact answer to a query



PostGIS – GIST indexes

GiST stands for "Generalized Search Tree" and is a generic form of indexing

PostGIS uses an R-Tree index implemented on top of GiST to index GIS data

GiST is used to speed up searches on all kinds of irregular data structures (integer arrays, spectral data, etc) which are not amenable to normal B-Tree indexing

```
CREATE INDEX [indexname] ON [tablename] USING GIST ( [geometryfield] );
```

Demonstration

Q1: What are the municipalities with cycleways?

Q2: What is the total length (in kilometers) of bike lanes in each municipality in Portugal?

Download and import source data

Get source data...

OSM → <https://download.geofabrik.de/europe/portugal.html>
(road network)

INE (Census 2021) → <https://mapas.ine.pt/download/index2011.phtml>
(Census 2021 statistics by subsection and geographical data)

INE (Administrative division codes) → <https://smi.ine.pt/Categoria>
Links the codes of the districts, municipalities and parishes with the corresponding names. See View Categories, Tabular View. Notice that Islands are not included.

Shapefile: can be imported using
shp2pgsql

GeoPackage file: can be imported using
QGIS using DB manager or the
processing toolbox

CSV file: use \COPY in pgsql, pgAdmin
import/export (GUI), Python, etc.

The source data



BGRI11_CONT

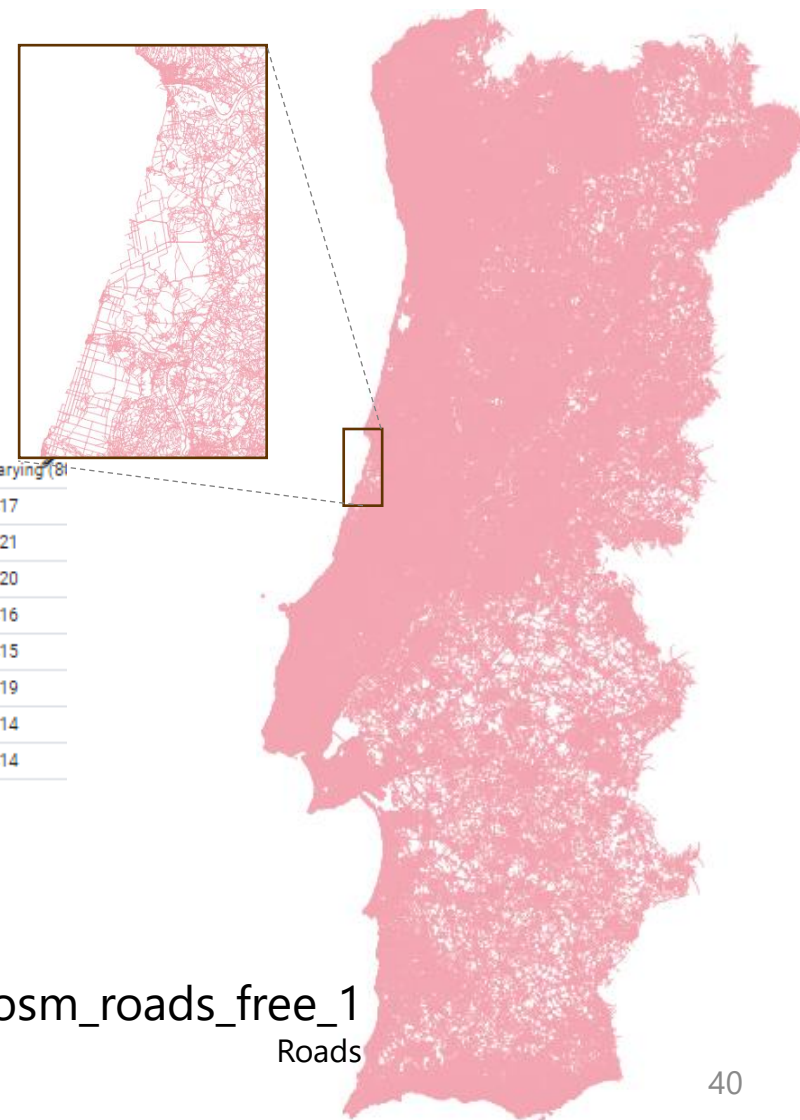
The geometries are subsections

DTMN11	FR11	SEC11	SS11	BGRI11	LUG11	LUG11DESIG	GEO_COD
character varying (8)	character varying (17)	character varying (4)	character varying (17)	character varying (16)	character varying (10)	character varying (90)	character varying (8)
0410	17	004	17	04101700417	013180	Vila Flor	04101700417
0410	17	005	21	04101700521	013180	Vila Flor	04101700521
0410	17	005	20	04101700520	013180	Vila Flor	04101700520
0410	17	004	16	04101700416	013180	Vila Flor	04101700416
0410	17	004	15	04101700415	013180	Vila Flor	04101700415
0410	17	005	19	04101700519	013180	Vila Flor	04101700519
0401	02	001	14	04010200114	013370	Alfândega da Fé	04010200114
0410	17	004	14	04101700414	013180	Vila Flor	04101700414

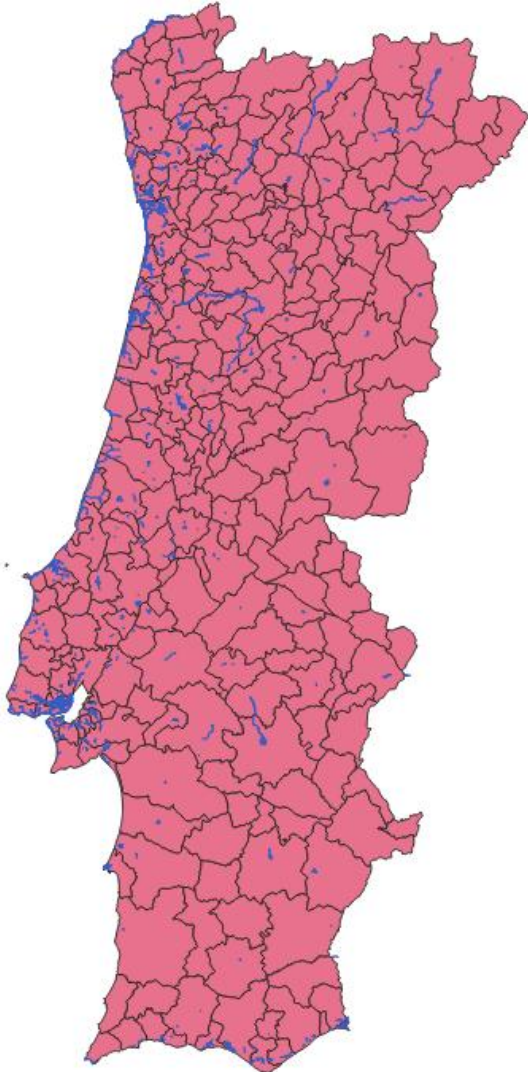
level	code	designation
integer	[PK] character varying (160)	character varying (160)
1	01	Aveiro
2	0101	Águeda
3	010103	Aguada de Cima
3	010109	Fermentelos
3	010112	Macinhata do Vouga

admincodes

gis_osm_roads_free_1
Roads



Transform the data to make it an easier to use



CREATE TABLE municipalities AS

```
SELECT  A.code, A.designation, ST_UNION(B.geom) as geom
FROM    admincodes A INNER JOIN "BGRI11_CONT" B ON A.code = B."DTMN11"
GROUP BY A.code
```

code character varying (4)	designation character varying (160)	st_astext text
0101	Águeda	POLYGON((-31931.045400000177 98955.66609999991,-31933.2953000000347 98937.94
0102	Albergaria-a-Velha	POLYGON((-33493.279600000001 104598.450200000078,-33564.519399999964 104624.6
0103	Anadia	POLYGON((-37583.172600000165 81805.497800000001,-37692.3706999999854 82229.3
0104	Arouca	POLYGON((-18825.0223000000302 136395.647800000038,-18834.0483999999738 13640
0105	Aveiro	POLYGON((-44234.016499999991 104997.620600000002,-44240.326399999967 105002.3
0106	Castelo de Paiva	POLYGON((-19837.6374000000356 148229.50819999991,-19839.6070999999674 148227
0107	Espinho	POLYGON((-42320.86469999998 144259.421299999958,-42389.6672999999855 144251.7
0108	Estarreja	POLYGON((-41741.891599999974 119955.21629999995,-41741.541199999986 119956.53
0109	Santa Maria da Feira	POLYGON((-38785.0580000000194 139323.804500000055,-38819.27429999997 139368.6
0110	Ílhavo	POLYGON((-51761.955900000041 101187.498999999984,-52088.564799999982 101286.1

Municipalities with cycleways

```
SELECT  DISTINCT M.designation
FROM    municipalities M, gis_osm_roads_free_1 R
WHERE   R.fclass = 'cycleway'
AND     ST_intersects(M.geom, ST_transform(R.geom, 3763))
```

- Execution time before creating spatial indexes: **14 secs 962 msec.**

- Creating spatial indexes

```
CREATE INDEX idx_osm_roads_geom ON gis_osm_roads_free_1 USING GIST (geom);
```

8 secs 707 msec.

```
CREATE INDEX idx_municipalities_geom ON municipalities USING GIST (geom);
```

30msec.

- Execution time after creating the spatial indexes: **843 msec.**

Length of cycleways by municipality

```
SELECT    M.designation,  
          SUM(ST_length(ST_Intersection(M.geom, ST_Transform(R.geom, 3763)))/1000) as "Length in km"  
FROM      municipalities M, gis_osm_roads_free_1 R  
WHERE     R.geom IS NOT NULL AND R.fclass = 'cycleway'  
GROUP BY M.designation
```

- Execution time with spatial indexes: **62 secs.**
- **ST_Intersection** is a spatial function with high computational cost but it cannot be avoided. **ST_Transform** has a non-negligible cost, but it is significantly lower than the previous one. A possible optimization:

```
ALTER TABLE gis_osm_roads_free_1 ADD COLUMN geom_3763 geometry(Geometry, 3763);
```

```
UPDATE gis_osm_roads_free_1 SET geom_3763 = ST_Transform(geom, 3763);
```

```
CREATE INDEX idx_roads_geom_3763 ON gis_osm_roads_free_1 USING GIST(geom_3763);
```

```
SELECT    M.designation,  
          SUM(ST_length(ST_Intersection(M.geom, R.geom_3763)))/1000) as "Length in km"    ...
```

- Execution time after SRID transformation: **57 secs.**

Bibliography

Nikos Mamoulis. Spatial data management. Synthesis Lectures on Data Management, Morgan&Claypool publishers, 2012.

- **Chapter 2 and 3**

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